CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNJAB

DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

Embedded Systems and Internet of Things Laboratory

Class Group G1 | Section A | Project Group 02

Project Abstract

Project Title: Campus Air quality, Carbon footprint, temperature monitoring using IOT

Abstract: This project focuses on monitoring campus environmental conditions by measuring air quality, carbon footprint, and temperature using IoT technology. Various sensors are deployed to continuously collect data related to air pollutants, carbon emission levels, and ambient temperature across campus. The collected data is processed using a microcontroller and transmitted to a cloud or monitoring platform for real-time visualization and analysis. This system helps in identifying pollution-prone areas and understanding the impact of human activities on the campus environment. The proposed solution is cost-effective, easy to implement, and suitable for educational institutions aiming to promote environmental awareness and sustainability.

Application Area:

- Campus Environmental Monitoring
- Pollution Analysis and Control
- Smart and Sustainable Campuses

Impact on Society/Industry:

- Healthier Living and Working Environment: Continuous monitoring helps maintain better air quality.
- Reduction of Carbon Footprint: Helps reduce carbon footprint through data-based monitoring.
- Support for Sustainable Infrastructure: Supports sustainable and smart campus or industrial environments.

Sustainable Development Goals: SDG 3 – Good Health and Well-Being, SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities, SDG 13 – Climate Action.

Technology Stack: ESP32, IoT, Environmental Sensors, Embedded C++, Cloud/IoT Platform

Group Details:

Sr. No	Name of Students	Roll No	Project Guide Name and Signature
1.	Abhay Singh Pundir	2410990006	Dr. Manvinder Sharma (ES&IOT Lab Faculty Name)
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Approved By:

Name of Faculty in Panel	Designation	Signature
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